Saravanan S

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Professional Summary

Motivated Electronics and Communication Engineering student with a strong interest in Embedded Systems, IoT, and Wireless Communication. Hands-on experience in developing microcontroller-based systems using Arduino Nano and ESP32, with practical knowledge of sensor interfacing, wireless communication, and real-time monitoring. Developed a disaster-resilient emergency communication system using nRF24L01 and currently developing an IoT-based smart waste management system using ESP32 and ultrasonic sensors. Eager to begin a career in Embedded Systems, IoT, or Electronics Engineering.

Technical Skills

Programming: Embedded C, C
Microcontrollers: Arduino Nano, ESP32
Sensors: HC-SR04 Ultrasonic Sensor
Modules & Components: nRF24L01, 16x2 I2C LCD, Push Buttons
Embedded Systems: Sensor Interfacing, GPIO, Microcontroller Programming, Real-Time Monitoring
Communication: I2C, Wi-Fi, Wireless Communication
IoT: IoT-Based Monitoring, Sensor Data Acquisition, Wireless Data Transmission
Tools: Arduino IDE, Visual Studio Code, Git, GitHub
Soft Skills: Problem Solving, Teamwork, Communication, Time Management

Internship Experience

KioTech Dec. 2025
Web Development Intern
- Gained practical experience with HTML, CSS, and JavaScript through web development exercises.
- Strengthened programming logic and problem-solving skills through practical development tasks.

Projects

Disaster-Resilient Emergency Communication Network | ***Arduino Nano, nRF24L01, I2C LCD***
- Developed a wireless emergency communication system using Arduino Nano and nRF24L01 modules for communication without internet or cellular networks.
- Implemented wireless transmission and reception of emergency alerts between multiple nodes using nRF24L01 modules.
- Integrated push buttons to trigger emergency alerts for flood, fire, and earthquake conditions.
- Displayed system status and emergency information using a 16x2 I2C LCD and assigned unique IDs to communication nodes.

Smart Waste Management System | ***ESP32, HC-SR04, IoT, Wi-Fi***
- Developing an IoT-based waste monitoring system using ESP32 and HC-SR04 ultrasonic sensors to measure waste-bin fill levels.
- Implemented ultrasonic sensor interfacing to determine the waste level inside bins.
- Using ESP32 Wi-Fi connectivity to transmit sensor data for real-time waste-bin monitoring.
- Developing a monitoring dashboard to display bin status and generate alerts when the waste level reaches a defined threshold.

Achievements

- Won First Prize in a Paper Presentation at Kongu Engineering College.
- Solved 100 DSA problems on LeetCode, strengthening programming and problem-solving skills.

Education

Excel Engineering College Aug. 2023 – Jul. 2027
Bachelor of Engineering in Electronics and Communication Engineering *CGPA: 7.7/10*

Certifications

- Responsive Web Design — L&T EduTech

Languages

English: Professional Working Proficiency
Tamil: Native Proficiency
Telugu: Conversational (Speaking & Understanding)